

PHASE 3 DEFENSE AND COMPARISON ANALYSIS REPORT



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OVERVIEW



Abstract

In this report, we document the final phase of a cybersecurity project focused on strengthening system defenses and evaluating the effectiveness of security measures. The objective was to apply firewall rules to secure a vulnerable system (Metasploitable3) and compare its exposure before and after the defenses. Using a simulated environment with Kali Linux and Metasploitable3, firewall configurations were implemented, tested, and validated through Nmap scanning and verification of closed ports.



Requirements

- Need #1: Identify open ports on Metasploitable3 using Nmap scanning.
- Need #2: Apply firewall rules (iptables) to block vulnerable services.
- Need #3: Verify applied rules by listing iptables configuration.
- Need #4: Compare Nmap scan results before and after defense implementation.



Deliverables

- Need #1: Screenshot of initial Nmap scan showing open ports.
- Need #2: Screenshot of iptables commands used to block ports.
- Need #3: Screenshot of iptables configuration confirming applied rules.
- Need #4: Screenshot of post-defense Nmap scan showing blocked/filtered ports.

TASK 1: SET UP METASPLOITABLE3 (VICTIM VM)



1A. Installed VirtualBox & Imported Metasploitable3

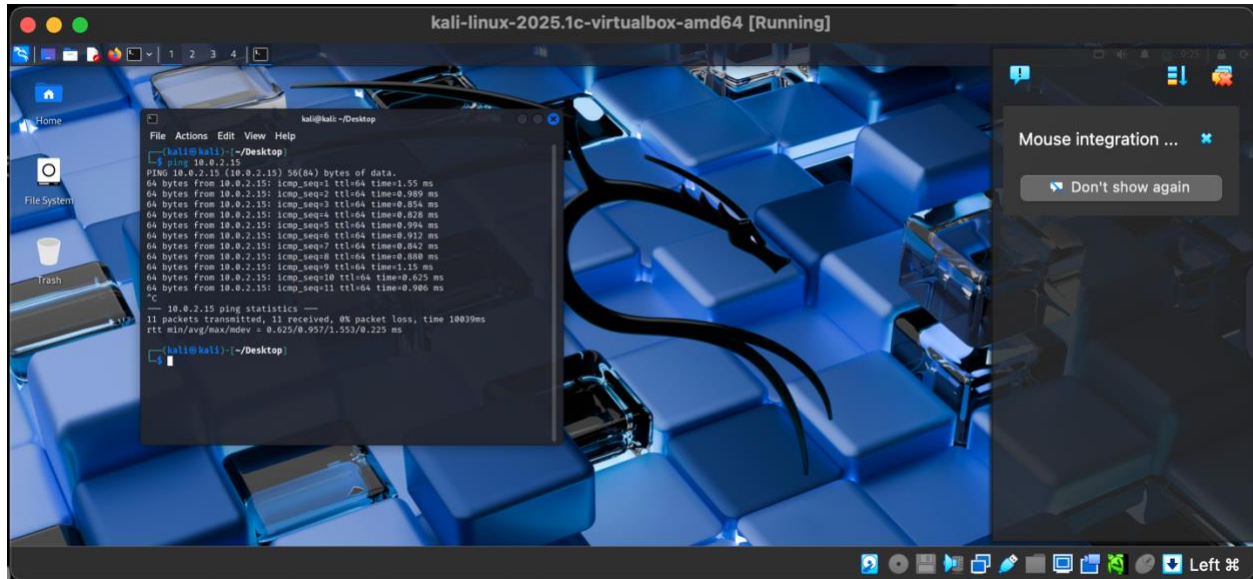


1B. Started and Logged into Metasploitable3

TASK 2: SET UP KALI LINUX (ATTACKER VM)



2A. Installed Kali Linux and Verified Setup



Verified connectivity between Kali and Metasploitable3 with successful ping.

TASK 3: INITIAL SCAN OF VULNERABLE SERVICES



3A. Ran Nmap Scan on Metasploitable3

ran nmap -A 10.0.2.15 to enumerate open services before applying defenses.

```
kali@kali: ~/Desktop
File Actions Edit View Help
(kali@kali)~[~/Desktop]
$ nmap -A 10.0.2.15
Starting Nmap 7.95 (https://nmap.org) at 2025-05-02 09:27 EDT
Nmap scan report for 10.0.2.15
Host is up (0.0011s latency).
Not shown: 991 filtered tcp ports (no-response)
PORT      STATE SERVICE VERSION
21/tcp    open  ftp      ProFTPD 1.3.5
22/tcp    open  ssh      OpenSSH 6.6.1p1 Ubuntu 2ubuntu2.13 (Ubuntu Linux; protocol 2.0)
|_ ssh-hostkey:
|_ 1024 2b:2e:1f:a4:54:26:87:76:12:26:59:58:0d:da:3b:04 (DSA)
|_ 2048 c9:ac:70:ef:f8:de:8b:a3:a3:44:ab:3d:32:0a:5c:6a (RSA)
|_ 256 c0:49:cc:18:7b:27:a4:07:0d:2a:0d:bb:42:4c:36:17 (ECDSA)
|_ 256 a0:76:f3:76:f8:f0:70:4d:09:ca:e1:10:fd:a9:cc:0a (ED25519)
80/tcp    open  http     Apache httpd 2.4.7
|_ http-server-header: Apache/2.4.7 (Ubuntu)
|_ http-1.1:
|_ http-1.1: Volume /
|_ http-1.1:
|_ http-1.1: FILENAME
|_ http-1.1: chat/
|_ http-1.1: 2011-07-27 20:17 drupal/
|_ http-1.1: 1.7K 2020-10-29 19:37 payroll_app.php
|_ http-1.1: 2013-04-08 12:06 phpmyadmin/
|_ http-1.1: Index of /
445/tcp    open  smb      Samba smbd 4.3.11-Ubuntu (workgroup: WORKGROUP)
631/tcp    open  ipp      CUPS 1.7
|_ http-robots.txt: 1 disallowed entry
|_ http-1.1:
|_ http-server-header: CUPS/1.7 IPP/2.1
|_ http-1.1: Home - CUPS 1.7.2
|_ http-1.1:
|_ http-1.1: Methods:
|_ http-1.1: PUT
|_ http-1.1: Potentially risky methods: PUT
3000/tcp   closed ppp
3306/tcp   open  mysql    MySQL (unauthorized)
8080/tcp   open  http     Jetty 8.1.7.v20120910
|_ http-server-header: Jetty(8.1.7.v20120910)
|_ http-1.1: Error 404 - Not Found
8191/tcp   closed intermapper
MAC Address: 08:00:27:16:93:0A (PCS Systemtechnik/Oracle VirtualBox virtual N IC)
Aggressive OS guesses: Linux 3.2 - 4.14 (98%), Linux 3.8 - 3.16 (98%), Linux 3.10 - 4.11 (94%), Linux 3.13 - 4.4 (94%), Linux 3.13 (94%), Linux 3.13 - 3.16 (94%), OpenWrt Chaos Calmer 15.05 (Linux 3.18) or Designated Driver (Linux 4.1 or 4.4) (94%), Linux 4.10 (94%), Android 5.0 - 6.0.1 (Linux 3.4) (94%), Android 8 - 9 (Linux 3.18 - 4.4) (94%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 1 hop
Service Info: Hosts: 127.0.2.1, METASPLOITABLE3-UB1404; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Host script results:
|_ clock-skew: mean: 1s, deviation: 2s, median: 0s
|_ smb-security-mode:
|_ account_used: guest
|_ authentication_level: user
|_ challenge_response: supported
|_ message_signing: disabled (dangerous, but default)
|_ smb-os-discovery:
|_ OS: Windows 6.1 (Samba 4.3.11-Ubuntu)
|_ Computer name: metasploitable3-ub1404
|_ NetBIOS computer name: METASPLOITABLE3-UB1404\x00
|_ Domain name: \x00
|_ FQDN: metasploitable3-ub1404
|_ System time: 2025-05-02T13:27:27+00:00
|_ smb2-time:
|_ date: 2025-05-02T13:27:28
|_ start_date: N/A
|_ smb2-security-mode:
|_ 3.1:1:
|_ Message signing enabled but not required

TRACEROUTE
HOP RTT ADDRESS
1 1.11 ms 10.0.2.15

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/.
Nmap done: 1 IP address (1 host up) scanned in 55.43 seconds
(kali@kali)~[~/Desktop]
$
```

Initial Nmap scan results showing open ports and services before firewall rules.

TASK 4: APPLY FIREWALL RULES WITH IPTABLES



4A. *Applied iptables Rules to Block Selected Services*

used iptables to block incoming traffic to ports 21 (FTP), 22 (SSH), and 3306 (MySQL) to harden the victim machine.

Commands used:

```
sudo iptables -A INPUT -p tcp --dport 21 -j DROP
```

```
sudo iptables -A INPUT -p tcp --dport 22 -j DROP
```

```
sudo iptables -A INPUT -p tcp --dport 3306 -j DROP
```

```
vagrant@metasploitable3-ub1404:~$ sudo iptables -A INPUT -p tcp --dport 21 -j DR  
OP  
vagrant@metasploitable3-ub1404:~$ _
```

Executed iptables rule to block FTP port 21.

TASK 5: VERIFY FIREWALL WITH IPTABLES LIST



5A. Verified Rules with iptables -L --line-numbers

```
18 1048 ACCEPT tcp -- * * 0.0.0.0/0 0.0.0.0/0
    tcp dpt:3306
82 4824 ACCEPT tcp -- * * 0.0.0.0/0 0.0.0.0/0
    tcp dpt:21
0 0 ACCEPT tcp -- * * 0.0.0.0/0 0.0.0.0/0
    tcp dpt:3500
20 1168 ACCEPT tcp -- * * 0.0.0.0/0 0.0.0.0/0
    tcp dpt:445
3 132 ACCEPT tcp -- * * 0.0.0.0/0 0.0.0.0/0
    tcp dpt:8181
0 0 ACCEPT tcp -- * * 0.0.0.0/0 0.0.0.0/0
    tcp dpt:6697
4270 221K DROP all -- * * 0.0.0.0/0 0.0.0.0/0
0 0 DROP tcp -- * * 0.0.0.0/0 0.0.0.0/0
    tcp dpt:21

Chain FORWARD (policy ACCEPT 0 packets, 0 bytes)
pkts bytes target prot opt in out source destination
0 0 DROP tcp -- * * 0.0.0.0/0 0.0.0.0/0
    tcp dpt:8989

Chain OUTPUT (policy ACCEPT 821 packets, 174K bytes)
pkts bytes target prot opt in out source destination
12731 3539K ACCEPT all -- * lo 0.0.0.0/0 0.0.0.0/0

vagrant@metasploitable3-ub1404:~$
```

Verified iptables rules list to confirm firewall configuration.

TASK 6: PERFORM NMAP SCAN AFTER FIREWALL



6A. Ran Nmap Again to Confirm Blocked Ports

reran nmap -A 10.0.2.15 after applying firewall rules to confirm that selected ports were successfully filtered.

Post-firewall Nmap scan showing remaining open ports after applying initial firewall rules.

```
kali@kali: ~/Desktop
File Actions Edit View Help
tps://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 55.43 seconds

(kali@kali) [~/Desktop]
$ nmap -A 10.0.2.15
Starting Nmap 7.95 ( https://nmap.org ) at 2025-05-02 09:56 EDT
Nmap scan report for 10.0.2.15
Host is up (0.00085s latency).
Not shown: 991 filtered tcp ports (no-response)
PORT      STATE SERVICE VERSION
21/tcp    open  ftp      ProFTPD 1.3.5
22/tcp    open  ssh      OpenSSH 6.6.1p1 Ubuntu 2ubuntu2.13 (Ubuntu Linux; protocol 2.0)
| ssh-hostkey:
| 1024 2b:2e:1f:a6:54:26:87:76:12:26:59:58:0d:da:3b:04 (DSA)
| 2048 c0:ac:70:ef:f8:de:8b:a3:d3:44:ab:3d:32:0a:5c:0a (RSA)
| 256 c0:49:cc:18:7b:27:a4:07:0d:2a:0d:bb:42:4c:36:17 (ECDSA)
| 256 a0:76:f3:76:f8:f0:70:4d:09:ca:e1:10:fd:a9:cc:0a (ED25519)
80/tcp    open  http     Apache httpd 2.4.7
|_ http-server-header: Apache/2.4.7 (Ubuntu)
|_ http-title: Index of /
|_ http-ls: Volume /
|_  SIZE TIME FILENAME
|_ - 2020-10-29 19:37 chat/
|_ - 2011-07-27 20:17 drupal/
|_ 1.7K 2020-10-29 19:37 payroll_app.php
|_ - 2013-04-08 12:06 phpmyadmin/
445/tcp   open  netbios-ssn Samba smbd 4.3.11-Ubuntu (workgroup: WORKGROUP)
631/tcp   open  http     CUPS 1.7
|_ http-robots.txt: 1 disallowed entry
|_ /
|_ http-server-header: CUPS/1.7 IPP/2.1
|_ http-title: Home - CUPS 1.7.2
|_ http-methods:
|_ - Potentially risky methods: PUT
3000/tcp  closed pop
3306/tcp  open  mysql    MySQL (unauthorized)
8080/tcp  open  http     Jetty 8.1.7.v20120910
|_ http-title: Error 404 - Not Found
|_ http-server-header: Jetty(8.1.7.v20120910)
8181/tcp  closed intermapper
MAC Address: 08:00:27:16:93:0A (PCS Systemtechnik/Oracle VirtualBox virtual N
TC)
Aggressive OS guesses: Linux 3.2 - 4.14 (98%), Linux 3.8 - 3.16 (98%), Linux
3.10 - 4.11 (94%), Linux 3.13 - 4.4 (94%), Linux 3.13 (94%), Linux 3.13 - 3.1
6 (94%), OpenWrt Chaos Calmer 15.05 (Linux 3.18) or Designated Driver (Linux
4.1 or 4.4) (94%), Linux 4.10 (94%), Android 5.0 - 6.0.1 (Linux 3.4) (94%), A
ndroid 8 - 9 (Linux 3.18 - 4.4) (94%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 1 hop
Service Info: Hosts: 127.0.2.1, METASPLOITABLE3-UB1404; OSs: Unix, Linux; CPE
: cpe:/o:linux:linux_kernel

Host script results:
| smb2-time:
|_ date: 2025-05-02T13:56:32
|_ start date: N/A
|_ smb2-security-mode:
|_ 3:1:1:
|_ Message signing enabled but not required
|_ smb-security-mode:
|_ account_used: guest
|_ authentication_level: user
|_ challenge_response: supported
|_ message_signing: disabled (dangerous, but default)
|_ clock-skew: mean: 0s, deviation: 2s, median: -1s
|_ smb-os-discovery:
|_ OS: Windows 6.1 (Samba 4.3.11-Ubuntu)
|_ Computer name: metasploitable3-ub1404
|_ NetBIOS computer name: METASPLOITABLE3-UB1404\*00
|_ Domain name: \*00
|_ FQDN: metasploitable3-ub1404
|_ System time: 2025-05-02T13:56:33+00:00

TRACEROUTE
HOP RTT ADDRESS
1 0.86 ms 10.0.2.15

OS and Service detection performed. Please report any incorrect results at ht
tps://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 55.37 seconds

(kali@kali) [~/Desktop]
$
```



```

5    ACCEPT    icmp -- anywhere          anywhere
6    ACCEPT    tcp  -- anywhere          anywhere      tcp dpt:ssh
7    ACCEPT    tcp  -- anywhere          anywhere      tcp dpt:http
8    ACCEPT    tcp  -- anywhere          anywhere      tcp dpt:3000
9    ACCEPT    tcp  -- anywhere          anywhere      tcp dpt:http
10   ACCEPT    tcp  -- anywhere          anywhere      tcp dpt:ipp
11   ACCEPT    tcp  -- anywhere          anywhere      tcp dpt:mysql
12   ACCEPT    tcp  -- anywhere          anywhere      tcp dpt:ftp
13   ACCEPT    tcp  -- anywhere          anywhere      tcp dpt:3500
14   ACCEPT    tcp  -- anywhere          anywhere      tcp dpt:micro
soft-ds
15   ACCEPT    tcp  -- anywhere          anywhere      tcp dpt:8181
16   ACCEPT    tcp  -- anywhere          anywhere      tcp dpt:6697
17   DROP      all  -- anywhere          anywhere
18   DROP      tcp  -- anywhere          anywhere      tcp dpt:ftp

Chain FORWARD (policy ACCEPT)
num target      prot opt source          destination
1    DROP        tcp  -- anywhere          anywhere      tcp dpt:8989

Chain OUTPUT (policy ACCEPT)
num target      prot opt source          destination
1    ACCEPT      all  -- anywhere          anywhere

vagrant@metasploitable3-ub1404:~$ sudo iptables -A INPUT -p tcp --dport 3306 -j
DROP
Bad argument '-'
Try `iptables -h' or 'iptables --help' for more information.
vagrant@metasploitable3-ub1404:~$ sudo iptables -A INPUT -p tcp --dport 3306 -j
DROP
vagrant@metasploitable3-ub1404:~$

```

Nmap scan verifying firewall configuration; filtered ports confirmed.

```
kali@kali: ~/Desktop
File Actions Edit View Help
| smb-security-mode:
|   account_used: guest
|   authentication_level: user
|   challenge_response: supported
|_ message_signing: disabled (dangerous, but default)
|_ clock-skew: mean: -2s, deviation: 2s, median: -3s

TRACEROUTE
HOP RTT ADDRESS
1 1.00 ms 10.0.2.15

OS and Service detection performed. Please report any incorrect results at ht
tps://nmap.org/submit/.
Nmap done: 1 IP address (1 host up) scanned in 55.17 seconds

(kali@kali) ~/Desktop
$ nmap -A 10.0.2.15
Starting Nmap 7.95 ( https://nmap.org ) at 2025-05-02 10:58 EDT
Nmap scan report for 10.0.2.15
Host is up (0.00097s latency).
Not shown: 994 filtered tcp ports (no-response)
PORT      STATE SERVICE VERSION
22/tcp    open  ssh      OpenSSH 6.6.1p1 Ubuntu 2ubuntu2.13 (Ubuntu Linux;
protocol 2.0)
| ssh-hostkey:
|   1024 2b:2e:1f:a4:54:26:87:76:12:26:59:58:0d:da:3b:04 (DSA)
|   2048 c9:ac:70:ef:f8:de:8b:a3:a3:44:ab:3d:32:0a:5c:6a (RSA)
|   256 c0:49:cc:18:7b:27:a4:07:0d:2a:0d:bb:42:4c:36:17 (ECDSA)
|   256 a0:76:f3:76:f8:f0:70:4d:09:ca:e1:10:fd:a9:cc:0a (ED25519)
445/tcp    open  netbios-ssn Samba smbd 4.3.11-Ubuntu (workgroup: WORKGROUP)
631/tcp    open  ipp       CUPS 1.7
|_ http-server-header: CUPS/1.7 IPP/2.1
|_ http-robots.txt: 1 disallowed entry
|_/
|_ http-title: Home - CUPS 1.7.2
|_ http-methods:
|   Potentially risky methods: PUT
3000/tcp   closed ppp
8080/tcp   open  http      Jetty 8.1.7.v20120910
|_ http-server-header: Jetty(8.1.7.v20120910)
|_ http-title: Error 404 - Not Found
8181/tcp   closed intermapper
MAC Address: 08:00:27:16:93:0A (PCS Systemtechnik/Oracle VirtualBox virtual N
IC)
Aggressive OS guesses: Linux 3.2 - 4.14 (98%), Linux 3.8 - 3.16 (98%), Linux
3.10 - 4.11 (94%), Linux 3.13 - 4.4 (94%), Linux 3.13 (94%), Linux 3.13 - 3.1
6 (94%), OpenWrt Chaos Calmer 15.05 (Linux 3.18) or Designated Driver (Linux
4.1 or 4.4) (94%), Linux 4.10 (94%), Android 5.0 - 6.0.1 (Linux 3.4) (94%), A
ndroid 8 - 9 (Linux 3.18 - 4.4) (94%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 1 hop
Service Info: Host: METASPLOITABLE3-UB1404; OS: Linux; CPE: cpe:/o:linux:linu
x_kernel

Host script results:
| smb-os-discovery:
|   OS: Windows 6.1 (Samba 4.3.11-Ubuntu)
|   Computer name: metasploitable3-ub1404
|   NetBIOS computer name: METASPLOITABLE3-UB1404\x00
|   Domain name: \x00
|   FQDN: metasploitable3-ub1404
|   System time: 2025-05-02T14:58:42+00:00
|_ smb2-security-mode:
|   3.1.1:
|_   Message signing enabled but not required
|_ smb-security-mode:
|   account_used: guest
|   authentication_level: user
|   challenge_response: supported
|_ message_signing: disabled (dangerous, but default)
|_ smb2-time:
|   date: 2025-05-02T14:58:41
|   start_date: N/A
|_ clock-skew: mean: -3s, deviation: 1s, median: -4s

TRACEROUTE
HOP RTT ADDRESS
1 0.97 ms 10.0.2.15

OS and Service detection performed. Please report any incorrect results at ht
tps://nmap.org/submit/.
Nmap done: 1 IP address (1 host up) scanned in 54.50 seconds

(kali@kali) ~/Desktop
$
```

Final Nmap scan confirming applied firewall rules; targeted ports successfully blocked.

TASK 7: FINAL ANALYSIS

Environment Setup:

- **Victim Environment:**
 -
 -
 - **System: Metasploitable3 – Ubuntu 14.04**
 - **Purpose: Vulnerable target system**
 - **IP Address: 10.0.2.15**
- **Attacker Environment:**
 - **System: Kali Linux**
 - **Tools Used: Nmap, iptables**

Defensive Focus:

- **Targeted Services: FTP (21), SSH (22), MySQL (3306)**
- **Defensive Action: Applied iptables firewall rules to block incoming connections on selected ports.**

Outcome:

- ✓ **Ports were accessible in initial scan.**
- ✓ **After firewall configuration, targeted ports were filtered/closed in Nmap.**
- ✓ **Firewall rules were confirmed using iptables -L.**

Conclusion:

This phase demonstrated how to configure basic firewall defenses on a vulnerable machine using iptables, verify configuration, and test effectiveness using scanning tools like Nmap. By successfully filtering ports, we reduced the attack surface of the system.